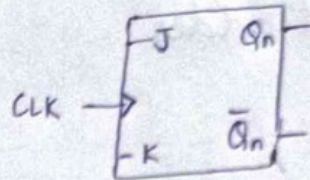


Q-1 Design JK flip flop and explain excitation table, truth table, characteristics equation and draw its logic circuit diagram. (C05)

JK flip flop is a improved version of SR flip flop



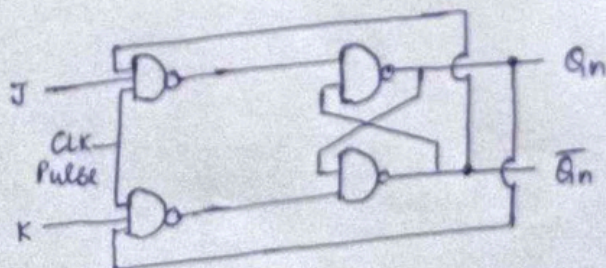
Truth Table.

J	K	Q_n (Present state)	Q_{n+1} (Next state)	Operation
0	0	0	0	Hold
0	0	1	1	
0	1	0	0	Reset
0	1	1	0	
1	0	0	1	Set
1	0	1	1	
1	1	0	1	Toggle
1	1	1	0	

Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Characteristic equation $\rightarrow Q_{n+1} = J\bar{Q}_n + Q_n\bar{K}$



Circuit Diagram

Q-2 Design and Implement Full Adder using Half adders. (CO4)

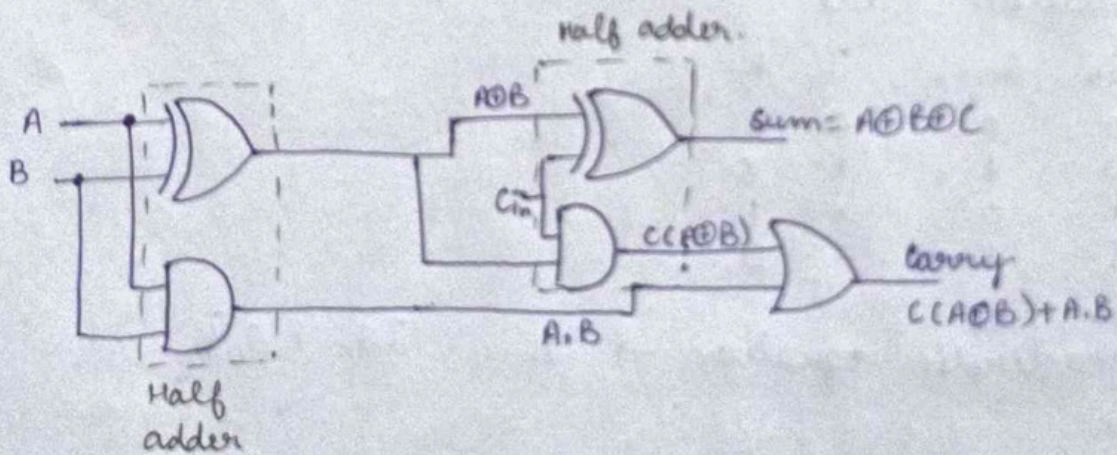
Expression for sum in Full adder

$$S = A \oplus B \oplus C$$

Expression for carry in Full adder

$$\text{Carry} = C(A \oplus B) + AB$$

A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Q-3. Simplify and realize the Boolean Expression using K-map: $F(A,B,C,D) = \sum m(0,2,5,6,8,11,13,14)$. and implement it using AOI. (05)

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$	1	0	0	1
$\bar{A}B$	0	1	0	1
AB	0	1	0	1
$A\bar{B}$	1	0	1	0

Group (0, 8) $\rightarrow \bar{B}\bar{C}\bar{D}$

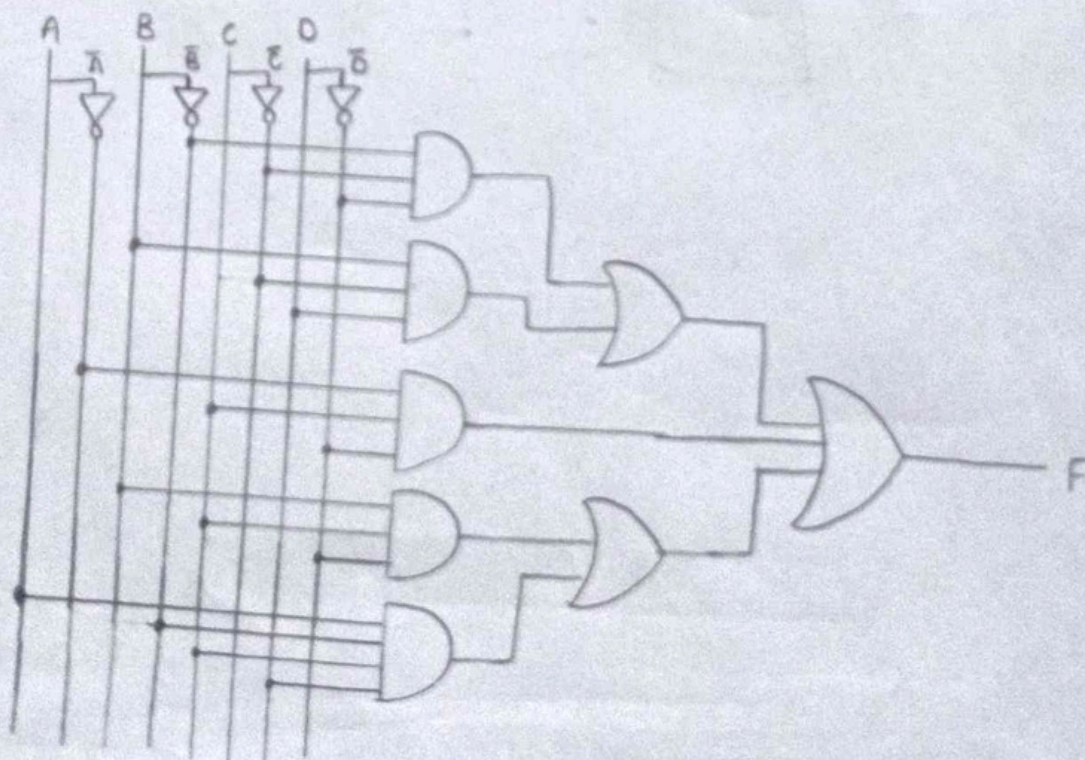
Group (5, 13) $\rightarrow B\bar{C}D$

Group (2, 6) $\rightarrow \bar{A}C\bar{D}$

Group (6, 14) $\rightarrow BC\bar{D}$

Group (11) $\rightarrow A\bar{B}CD$

$$F = \bar{B}\bar{C}\bar{D} + B\bar{C}D + \bar{A}C\bar{D} + BC\bar{D} + A\bar{B}CD$$



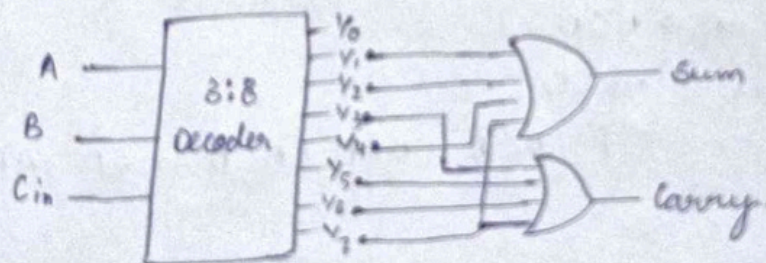
Q-4 Analyze any combinational circuit using 3:8 decoder. (CO4)
 Analyzing full adder using 3:8 decoder.

Truth Table of Full adder

A	B	C _{in}	Sum	Carry	
0	0	0	0	0	$Y_0 \rightarrow \bar{A}\bar{B}\bar{C}$
0	0	1	1	0	$Y_1 \rightarrow \bar{A}\bar{B}C$
0	1	0	1	0	$Y_2 \rightarrow \bar{A}B\bar{C}$
0	1	1	0	1	$Y_3 \rightarrow \bar{A}BC$
1	0	0	1	0	$Y_4 \rightarrow A\bar{B}\bar{C}$
1	0	1	0	1	$Y_5 \rightarrow A\bar{B}C$
1	1	0	0	1	$Y_6 \rightarrow AB\bar{C}$
1	1	1	1	1	$Y_7 \rightarrow ABC$

$$\text{Sum} = \sum m(1, 2, 4, 7) = Y_1 + Y_2 + Y_4 + Y_7$$

$$\text{Carry} = \sum m(3, 5, 6, 7) = Y_3 + Y_5 + Y_6 + Y_7$$



Q-5 Analyze the boolean function using 4:1 MUX
 $F(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 9, 12, 14)$. (CO4).

$$F(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 9, 12, 14)$$

For 4:1 MUX

2 select lines let say C, D.

4 input lines (I_0, I_1, I_2, I_3)

1 Output $\rightarrow F$

A	B	Minterms
0	0	$I_0 (0, 1, 4) \rightarrow A' + B'$
0	1	$I_1 (2, 6) \rightarrow A \oplus B$
1	0	$I_2 (9) \rightarrow AB'$
1	1	$I_3 (12, 14) \rightarrow AB$

